Exact modular-form certificates for four 2-colour overpartition congruences

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Abstract

For positive integers ℓ, define $\sum_{n\geq 0}\overline{p}_\ell(n)q^n = f_2 f_{2\ell}/(f_1^2 f_\ell^2)$, where $f_k = \prod_{m\geq 1}(1-q^{km})$. We give exact modular-form certificates for $\overline{p}_8(32n+28)\equiv 0 \pmod{32}$, $\overline{p}_6(24n+17)\equiv \overline{p}_6(24n+23)\equiv 0 \pmod{64}$ and $\overline{p}_9(36n+33)\equiv 0 \pmod{128}$. The first statement already occurs in Bharadwaj–Sujatha–Chandankumar's Theorem 4.4, as well as their conjecture list; we provide a complete independent certificate, not new statement priority. At most 541 extracted Fourier coefficients are tested against a Sturm bound. The largest selected index of the underlying form is 19,440, and the direct progression confirmation reaches 19,473.

1 Introduction

Let $f_k = (q^k;q^k)_\infty$. In [1] Bharadwaj, Sujatha and Chandankumar study the functions $\overline{p}_\ell(n)$ with

$$\sum_{n\geq 0}\overline{p}_\ell(n)q^n = \frac{f_2\, f_{2\ell}}{f_1^2\, f_\ell^2},$$

the generating function of pairs of overpartitions in which the second component has all parts divisible by ℓ. Their methods are elementary (q-series dissections), and they deliberately avoid modular forms. In their final section they state the following.

Theorem 1 (Statements in [1]; attribution below)**.** *For all $n \geq 0$,*

$$\begin{aligned}\overline{p}_8(32n+28) &\equiv 0 \pmod{32},\\ \overline{p}_6(24n+17) &\equiv 0 \pmod{64},\\ \overline{p}_6(24n+23) &\equiv 0 \pmod{64}.\end{aligned}$$

Theorem 2 (Second congruence of a further conjecture in [1])**.** *For all $n \geq 0$, $\overline{p}_9(36n+33)\equiv 0 \pmod{128}$.*

The first congruence is the $\alpha = 2$ case of [1, Theorem 4.4, equation (4.51)], and also appears in its Conjecture 2, equation (5.4). The inspected proof does not explicitly derive the full all-α assertion. We record this ambiguity rather than claim either new statement priority or a complete earlier proof. The other three are equations (5.5), (5.6), (5.8) of that source. No exact earlier proof was located in the bounded search documented with this collection; this is not a priority certification. The companion family paper treats the modulo-8 parameter family and extends the dyadic modulo-16 family.

We prove Theorems 1 and 2 by the standard method of Ono and Radu [2, 3]: we multiply the generating function by an eta-quotient in q^M chosen to make it a holomorphic modular form,

apply U_M, and check the finitely many coefficients that Sturm's theorem [4] requires. The only work is finding suitable multipliers. We found them by integer linear programming; the proof itself depends only on checking the certificates in Table 1.

2 Tools

Let $\eta(z) = q^{1/24} f_1$ with $q = e^{2\pi i z}$. An eta-quotient is $F(z) = \prod_{d|N} \eta(dz)^{r_d}$.

Lemma 3 (Gordon–Hughes–Newman, Ligozat; [2, Thms. 1.64, 1.65])**.** *Suppose $k = \frac{1}{2}\sum r_d$ is a positive integer, $\sum_d d\, r_d \equiv 0 \pmod{24}$ and $\sum_d (N/d)\, r_d \equiv 0 \pmod{24}$. Then F satisfies the transformation law of weight k on $\Gamma_0(N)$ with character $\chi(m) = (\frac{(-1)^k s}{m})$, where $s = \prod d^{r_d}$. If moreover for every divisor $c \mid N$*

$$\frac{N}{24} \sum_{d|N} \frac{\gcd(c,d)^2\, r_d}{\gcd(c, N/c)\, c\, d} \;\geq\; 0,$$

then F is holomorphic at all cusps, so $F \in M_k(\Gamma_0(N), \chi)$.

Lemma 4 ([2, Prop. 2.22])**.** *Let χ be a Dirichlet character modulo N. If $M \mid N$, then $U_M : \sum a(n) q^n \mapsto \sum a(Mn) q^n$ maps $M_k(\Gamma_0(N), \chi)$ to itself.*

Lemma 5 (Sturm [4]; [2, Thm. 2.58])**.** *Let $G \in M_k(\Gamma_0(N), \chi)$ have integer coefficients $a(n)$, and let p be a prime. If $a(n) \equiv 0 \pmod{p}$ for all $n \leq S = \frac{k}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, then $a(n) \equiv 0 \pmod{p}$ for all n.*

Applying Lemma 5 j times shows that if $2^j \mid a(n)$ for all $n \leq S$ then $2^j \mid G$. After each step, $G/2$ is again in $M_k(\Gamma_0(N), \chi)$ with integer coefficients.

3 The reduction

Fix ℓ, a modulus M and a residue r. Let $A_\ell(z) = \eta(2z)\eta(2\ell z)/(\eta(z)^2 \eta(\ell z)^2)$. Its exponents satisfy $\sum d\, r_d = 2 + 2\ell - 2 - 2\ell = 0$, so $A_\ell = \sum \overline{p}_\ell(n) q^n$ exactly. Choose integers b_d for $M \mid d \mid N$ and put

$$B(z) = \prod_{M|d|N} \eta(dz)^{b_d} = q^c\, g(q^M), \qquad c = \tfrac{1}{24}\textstyle\sum d\, b_d, \quad g \in 1 + q\mathbb{Z}[[q]].$$

Suppose $c \in \mathbb{Z}$ and $c \equiv -r \pmod{M}$, and that $F = A_\ell B$ satisfies the hypotheses of Lemma 3. Then

$$F = q^c g(q^M) \sum_{n \geq 0} \overline{p}_\ell(n) q^n,$$

and the exponents divisible by M are exactly those with $n \equiv r \pmod{M}$. Hence

$$G := F \mid U_M = q^{(c+r)/M}\, g(q) \sum_{n \geq 0} \overline{p}_\ell(Mn + r)\, q^n \in M_k(\Gamma_0(N), \chi)$$

by Lemma 4. Since g is a unit in $\mathbb{Z}[[q]]$, 2^j divides every coefficient of G if and only if $2^j \mid \overline{p}_\ell(Mn+r)$ for all $n \geq 0$. By the remark after Lemma 5, it suffices to check the coefficients of G up to q^S.

4 Certificates

Table 1 lists the multipliers. In each case:

- $\sum(N/d)r_d = 0$ for $F = A_\ell B$;
- the minimum over the cusps of Ligozat's order is 0, so the hypotheses of Lemma 3 hold;
- $M \mid N$ and χ is a character modulo N, so Lemma 4 applies (indeed the conductor of χ, 1 or 3, even divides N/M).

For (A), $s = 2^{156}$ is a square and k is even, so χ is trivial. In the other cases k is odd and the squarefree part of s is 3, so $\chi = (\frac{-3}{\cdot})$.

(ℓ, M, r)	N	k	c	S	j	coeffs
$(8, 32, 28)$	128	18	4	288	5	289
$(6, 24, 17)$	288	11	7	528	6	529
$(6, 24, 23)$	288	11	1	528	6	529
$(9, 36, 33)$	216	15	3	540	7	541

$$
\begin{aligned}
B_{8,28} &= \eta_{32}^{69}\eta_{64}^{-29}\eta_{128}^{-2},\\
B_{6,17} &= \eta_{24}^{61}\eta_{48}^{-31}\eta_{72}^{-16}\eta_{96}^{2}\eta_{144}^{8},\\
B_{6,23} &= \eta_{24}^{59}\eta_{48}^{-25}\eta_{72}^{-16}\eta_{96}^{-2}\eta_{144}^{8},\\
B_{9,33} &= \eta_{36}^{88}\eta_{72}^{-43}\eta_{108}^{-26}\eta_{216}^{13}.
\end{aligned}
$$

Table 1: Certificates; $\eta_d = \eta(dz)$. The indices $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$ are 192, 576, 432.

Proposition 6. *For each row of Table 1, all coefficients of $q^0, \dots, q^S$ in G are divisible by 2^j.*

Proof. This is an exact integer computation. The failure-sensitive script `verify.py` expands F, applies U_M, tests every coefficient $0 \le m \le S$, and checks the extraction identity and direct progression. A separate standard-library checker reconstructs the overpartition coefficients by theta inversion and the multiplier by a logarithmic-derivative recurrence, without FLINT. The historical auxiliary checker shared FLINT and a pentagonal-product approach and is not described as independent reproduction. A compact receipt records cusp orders, characters, shifts and exact ranges. The universal conclusion also requires the written modular-form argument above. □

Proof of Theorems 1 and 2. Combine Section 3 with Proposition 6. □

Remark 7. Explicit witnesses are $\overline{p}_8(28) = 86{,}368 = 32 \cdot 2699$, $\overline{p}_6(17) = 3648 = 64 \cdot 57$, $\overline{p}_6(23) = 25{,}024 = 64 \cdot 391$ and $\overline{p}_9(69) = 594{,}986{,}112 = 128 \cdot 4{,}648{,}329$. Thus none of the four moduli can be doubled uniformly. Stronger subprogressions are not excluded. The integer-programming discovery search is not a proof prerequisite and no minimality of its multipliers is claimed.

Version and assurance

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References

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